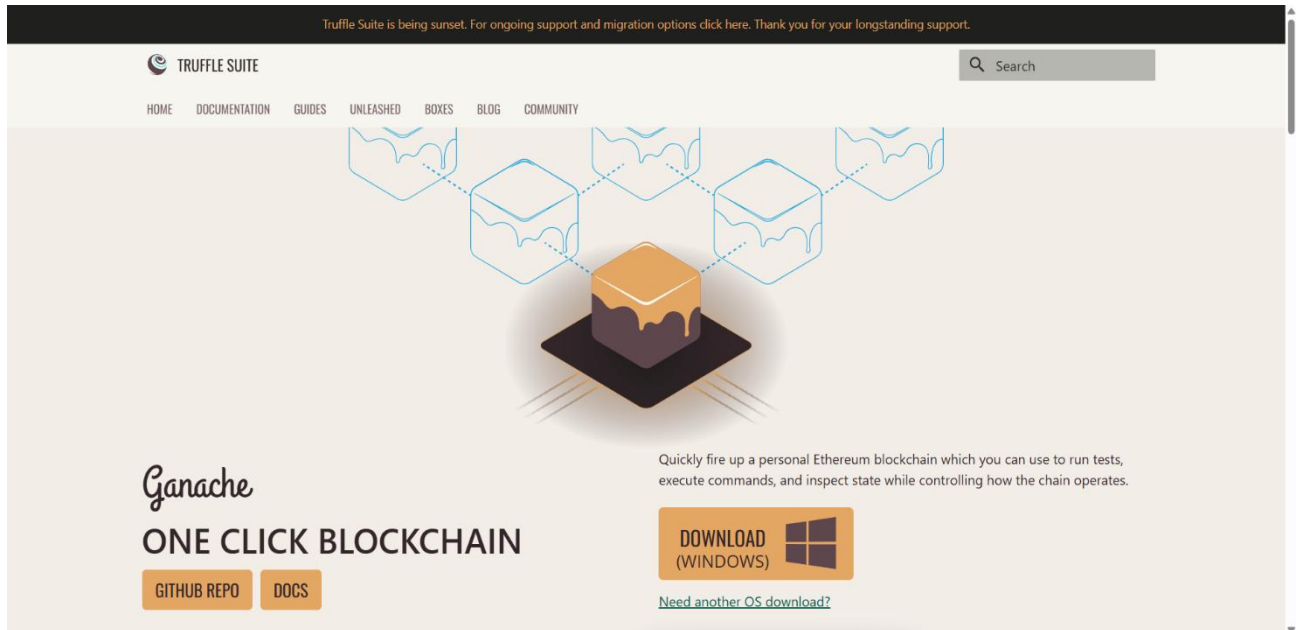


Experiment 6

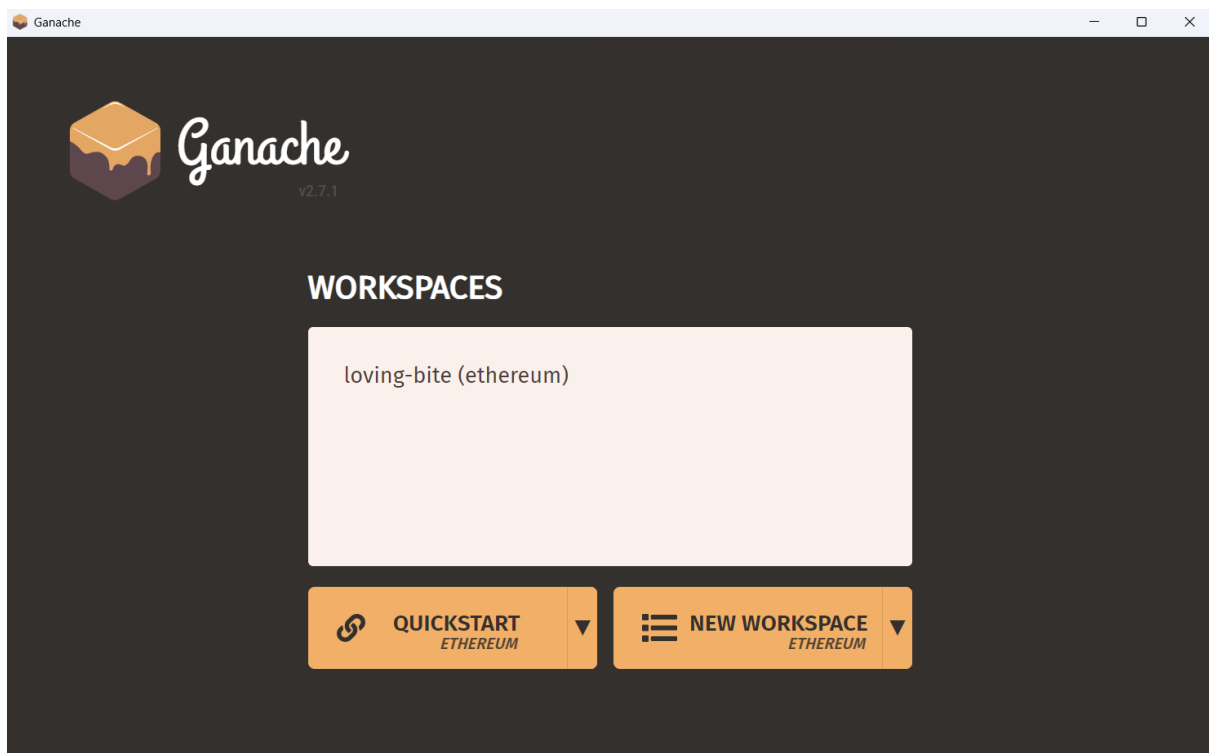
Aim: Blockchain platform Ethereum using Ganache.

Activity:

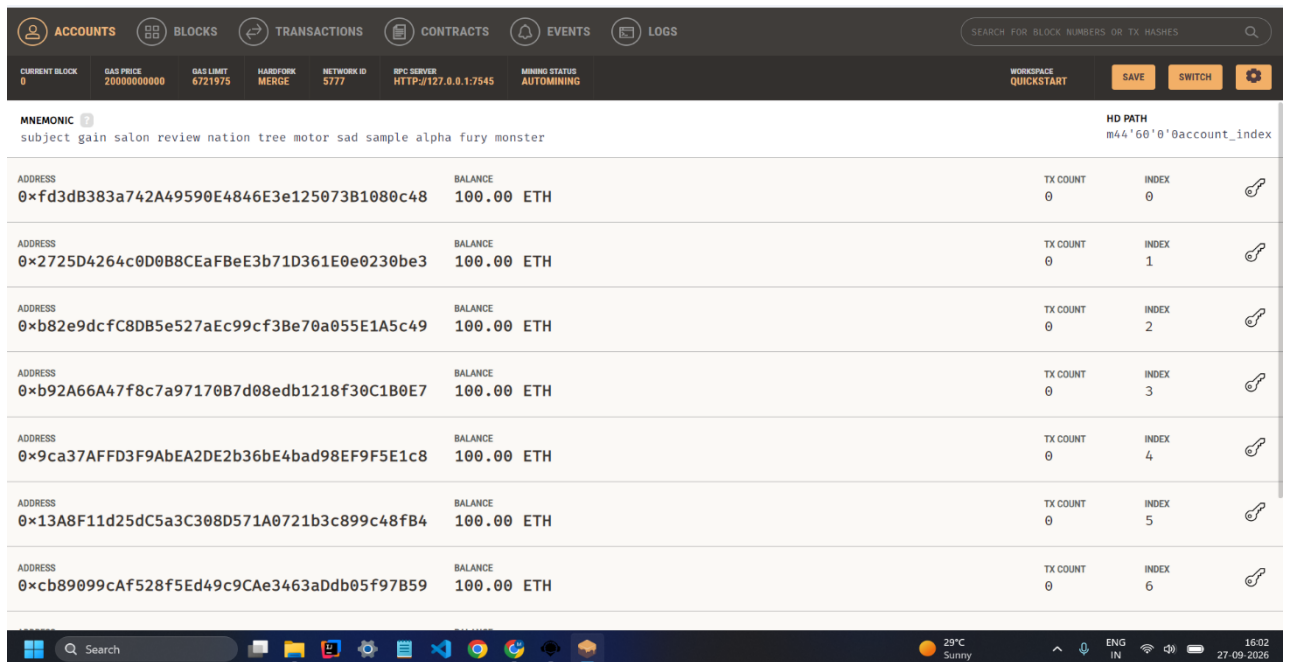
Step1: Install and setup Ganache



Step 2: Go to quick start and get ready to use setup



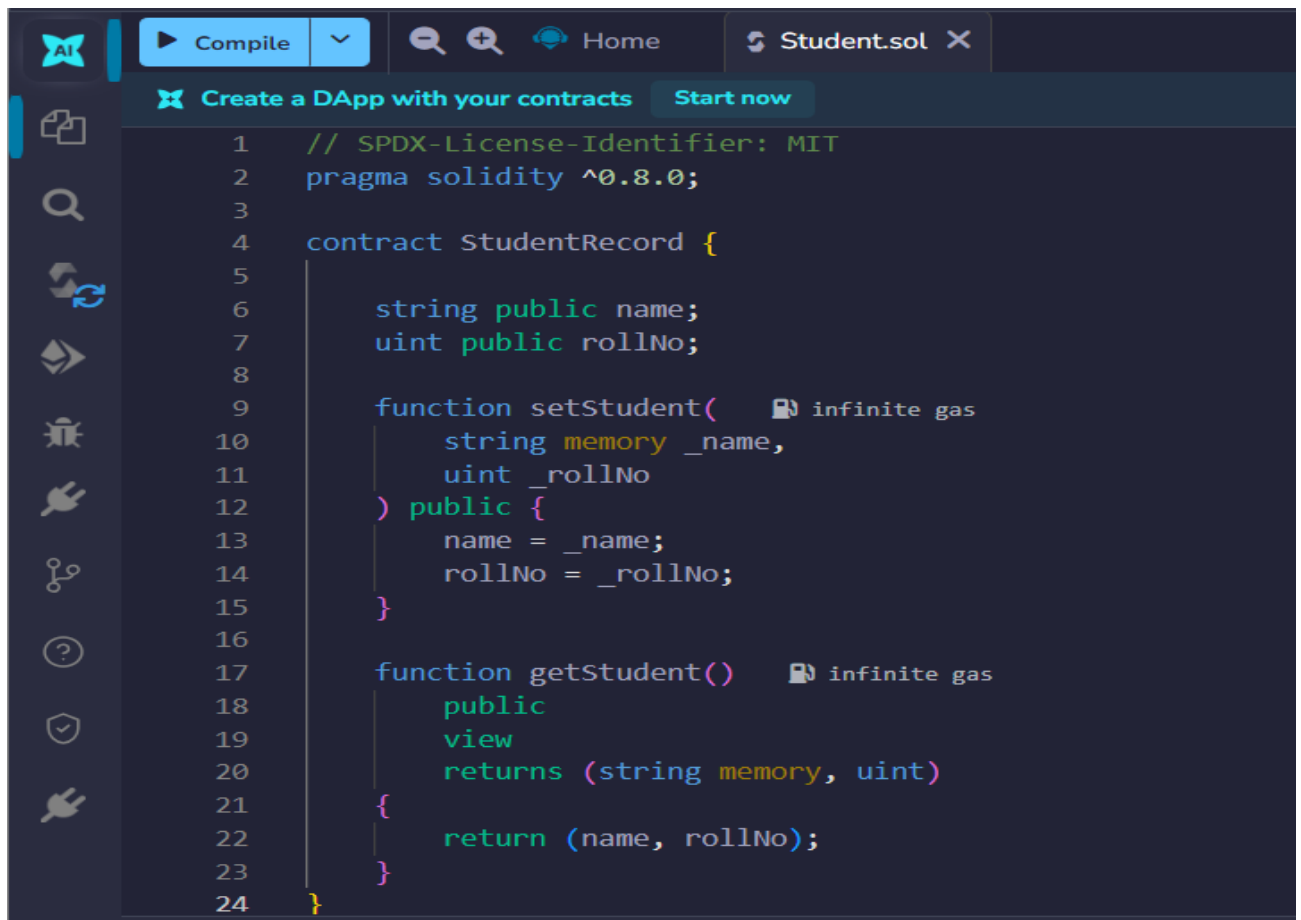
Step 3: Found the account credentials and RPC URL details in dashboard



The screenshot shows the Ethereum dashboard with the 'ACCOUNTS' tab selected. The top navigation bar includes links for ACCOUNTS, BLOCKS, TRANSACTIONS, CONTRACTS, EVENTS, and LOGS. A search bar is available for block numbers or transaction hashes. The dashboard displays various metrics: CURRENT BLOCK (0), GAS PRICE (2000000000), GAS LIMIT (6721975), HARDFORK (MERGE), NETWORK ID (5777), RPC SERVER (HTTP://127.0.0.1:7545), and MINING STATUS (AUTOMINING). A mnemonic phrase is shown: 'subject gain salon review nation tree motor sad sample alpha fury monster'. The HD PATH is 'm44'60'0'0account_index'. Below this, a table lists account details:

ADDRESS	BALANCE	TX COUNT	INDEX
0xfd3dB383a742A49590E4846E3e125073B1080c48	100.00 ETH	0	0
0x2725D4264c0D0B8CEaFBeE3b71D361E0e0230be3	100.00 ETH	0	1
0xb82e9dcfC8DB5e527aEc99cf3Be70a055E1A5c49	100.00 ETH	0	2
0xb92A66A47f8c7a97170B7d08edb1218f30C1B0E7	100.00 ETH	0	3
0x9ca37AFFD3F9AbEA2DE2b36bE4bad98EF9F5E1c8	100.00 ETH	0	4
0x13A8F11d25dC5a3C308D571A0721b3c899c48fB4	100.00 ETH	0	5
0xcb89099cAf528f5Ed49c9CAe3463aDdb05f97B59	100.00 ETH	0	6

Step 4: Open Remix IDE and create an contract



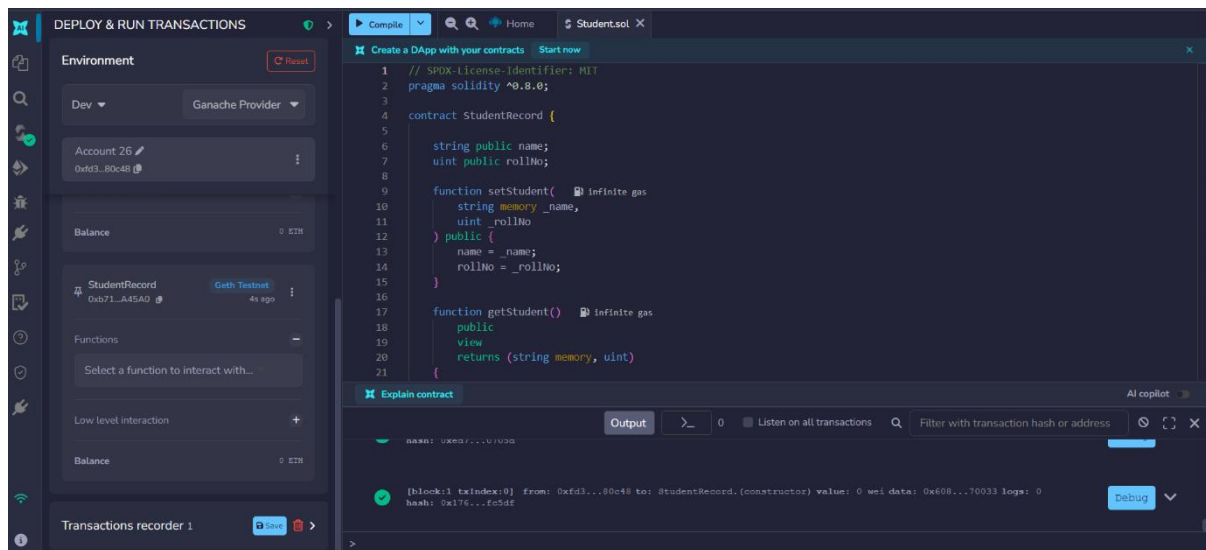
The screenshot shows the Remix IDE interface with a Solidity contract named 'StudentRecord' being created. The contract is defined as follows:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract StudentRecord {
5
6     string public name;
7     uint public rollNo;
8
9     function setStudent(    infinite gas
10         string memory _name,
11         uint _rollNo
12     ) public {
13         name = _name;
14         rollNo = _rollNo;
15     }
16
17     function getStudent()    infinite gas
18         public
19         view
20         returns (string memory, uint)
21     {
22         return (name, rollNo);
23     }
24 }
```

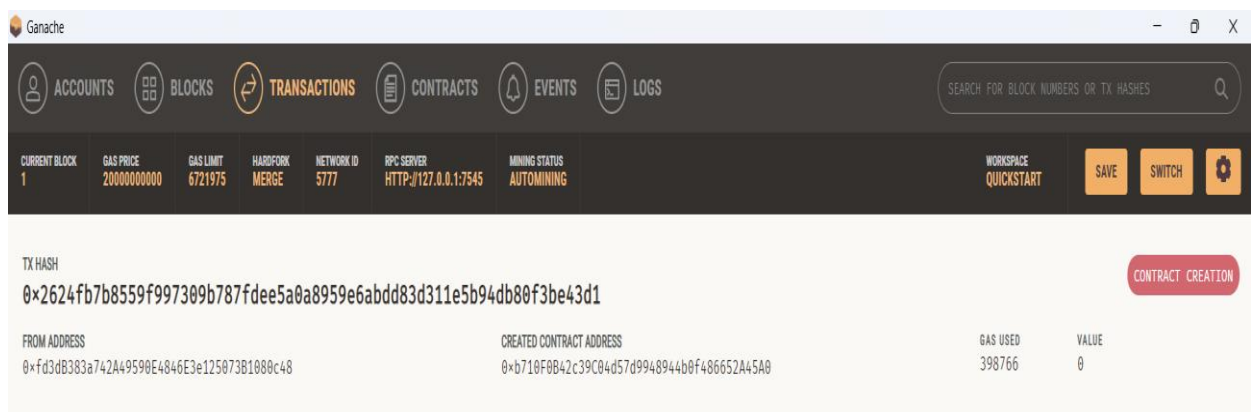
Step 5: Go to compile and deploy and use the RPC URL given in Ganache and for connecting with it

1. Select Environment Dev
2. Then provider as Ganache Provider
3. Then in URL use Ganache provided URL
4. Then click on Ok to connect with Ganache index 0 account

Step 6: Compile and deploy your contract using the RPC



Step 7: Use function and make transaction , See blocks for transaction blocks and transaction each transaction being done through IDE in Ganache



Step 8: Perform transaction and function call.

setStudent function call

1)

The screenshot shows the Ganache interface with the 'setStudent' function call transaction details. The left sidebar shows the 'Environment' tab with 'Dev' and 'Ganache Provider' selected. The main area displays the transaction details for 'setStudent' with parameters 'string, uint256'. The transaction hash is 0x8e4d349368624d7a2971ad3cb4fb6e70311bd951172e807fbc3e7f2149d0b2. The transaction cost is 67458 gas. The decoded input shows the function call parameters: 'string_name': 'Sunil', 'uint256_rollNo': '2303020'. The transaction status is '1 Transaction mined and execution completed'.

Environment

Dev Ganache Provider

Account 26 99.999 ETH

setStudent string, uint256

Sunil

2303020

Calldata Parameters

Value 0 wei

Gas Limit auto 0

Transact

Balance 0 ETH

Transactions recorder 2

Decoded output

Logs

Raw logs

[block:2 txIndex:0] from: 0xfd3...80c48 to: StudentRecord.setStudent(string,uint256) 0xb71...A45A0 value: 0 wei data: 0x9b4...00000 logs: 0 hash: 0x841...c61c0

status 1 Transaction mined and execution completed

transaction hash 0x8e4d349368624d7a2971ad3cb4fb6e70311bd951172e807fbc3e7f2149d0b2

block hash 0x841ac5a9f01bc2d1f2bcb0cb2f2f3c5706fe0e6327640bcb847a381e70c1c0

block number 2

from 0xfd3db383a742a49590e4846e3e125073b1080c48

to StudentRecord.setStudent(string,uint256) 0xb710f0b42c39c04d57d9948944b0f486652a45a0

transaction cost 67458 gas

decoded input { "string_name": "Sunil", "uint256_rollNo": "2303020" }

decoded output

logs

raw logs

The screenshot shows the Ganache interface with the 'TRANSACTIONS' tab selected. The top bar shows the current block number 2, gas price 20000000000, gas limit 6721975, hardfork MERGE, network ID 5777, RPC server HTTP://127.0.0.1:7545, and mining status AUTOMINING. The main area displays two transactions. The first transaction is a function call with TX HASH 0x8e4d349368624d7a2971ad3cb4fb6e70311bd951172e807fbc3e7f2149d0b2, FROM ADDRESS 0xfd3db383a742a49590e4846e3e125073b1080c48, TO CONTRACT ADDRESS 0xb710f0b42c39c04d57d9948944b0f486652a45a0, GAS USED 67458, and VALUE 0. The second transaction is a contract creation with TX HASH 0x2624fb7b8559f997309b787fdee5a0a8959e6abdd83d311e5b94db0f3be43d1, FROM ADDRESS 0xfd3db383a742a49590e4846e3e125073b1080c48, CREATED CONTRACT ADDRESS 0xb710f0b42c39c04d57d9948944b0f486652a45a0, GAS USED 398766, and VALUE 0.

Ganache

ACCOUNTS BLOCKS TRANSACTIONS CONTRACTS EVENTS LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK 2 GAS PRICE 20000000000 GAS LIMIT 6721975 HARDFORK MERGE NETWORK ID 5777 RPC SERVER HTTP://127.0.0.1:7545 MINING STATUS AUTOMINING

WORKSPACE QUICKSTART SAVE SWITCH

TX HASH 0x8e4d349368624d7a2971ad3cb4fb6e70311bd951172e807fbc3e7f2149d0b2 CONTRACT CALL

FROM ADDRESS 0xfd3db383a742a49590e4846e3e125073b1080c48 TO CONTRACT ADDRESS 0xb710f0b42c39c04d57d9948944b0f486652a45a0 GAS USED 67458 VALUE 0

TX HASH 0x2624fb7b8559f997309b787fdee5a0a8959e6abdd83d311e5b94db0f3be43d1 CONTRACT CREATION

FROM ADDRESS 0xfd3db383a742a49590e4846e3e125073b1080c48 CREATED CONTRACT ADDRESS 0xb710f0b42c39c04d57d9948944b0f486652a45a0 GAS USED 398766 VALUE 0

getStudent Function call

[illegible]

2)

[illegible]

EVENTS

NO EVENTS

NO EVENTS

Account with transaction count

CURRENT BLOCK5

GAS PRICE20000000000

GAS LIMIT6721975

HARDFORKMERGE

NETWORK ID5777

RPC SERVERHTTP://127.0.0.1:7545

MINING STATUSAUTOMINING

WORKSPACE

QUICKSTART

SAVE

SWITCH

MEMONIC

subject gain salon review nation tree motor sad sample alpha fury monster

HD PATH

m44'60'0'0account_index

ADDRESS	BALANCE	TX COUNT	INDEX	
0xfd3dB383a742A49590E4846E3e125073B1080c48	100.00 ETH	5	0	
ADDRESS	BALANCE	TX COUNT	INDEX	
0x2725D4264c0D0B8CEaFBeE3b71D361E0e0230be3	100.00 ETH	0	1	
ADDRESS	BALANCE	TX COUNT	INDEX	
0xb82e9dcfc8DB5e527aEc99cf3Be70a055E1A5c49	100.00 ETH	0	2	
ADDRESS	BALANCE	TX COUNT	INDEX	
0xb92A66A47f8c7a97170B7d08edb1218f30C1B0E7	100.00 ETH	0	3	
ADDRESS	BALANCE	TX COUNT	INDEX	
0x9ca37AFD3F9AbEA2DE2b36bE4bad98EF9F5E1c8	100.00 ETH	0	4	
ADDRESS	BALANCE	TX COUNT	INDEX	
0x13A8F11d25dC5a3C308D571A0721b3c899c48fB4	100.00 ETH	0	5	
ADDRESS	BALANCE	TX COUNT	INDEX	
0xcb89099cAf528f5Ed49c9CAe3463aDdb05f97B59	100.00 ETH	0	6	

Blocks

ACCOUNTS BLOCKS TRANSACTIONS CONTRACTS EVENTS LOGS					SEARCH FOR BLOCK NUMBERS OR TX HASHES	
CURRENT BLOCK GAS PRICE GAS LIMIT HARDFORK NETWORK ID RPC SERVER MINING STATUS					WORKSPACE LOVING-BITE SWITCH	
5 20000000000 6721975 MERGE 5777 HTTP://127.0.0.1:7545 AUTOMINING						
BLOCK	MINED ON		GAS USED		1 TRANSACTION	
5	2026-09-27 16:16:01		35503			
BLOCK	MINED ON		GAS USED		1 TRANSACTION	
4	2026-09-27 16:14:52		35539			
BLOCK	MINED ON		GAS USED		1 TRANSACTION	
3	2026-09-27 16:11:56		35515			
BLOCK	MINED ON		GAS USED		1 TRANSACTION	
2	2026-09-27 16:09:28		67458			
BLOCK	MINED ON		GAS USED		1 TRANSACTION	
1	2026-09-27 16:03:09		398766			
BLOCK	MINED ON		GAS USED		NO TRANSACTIONS	
0	2026-09-27 16:01:16		0			